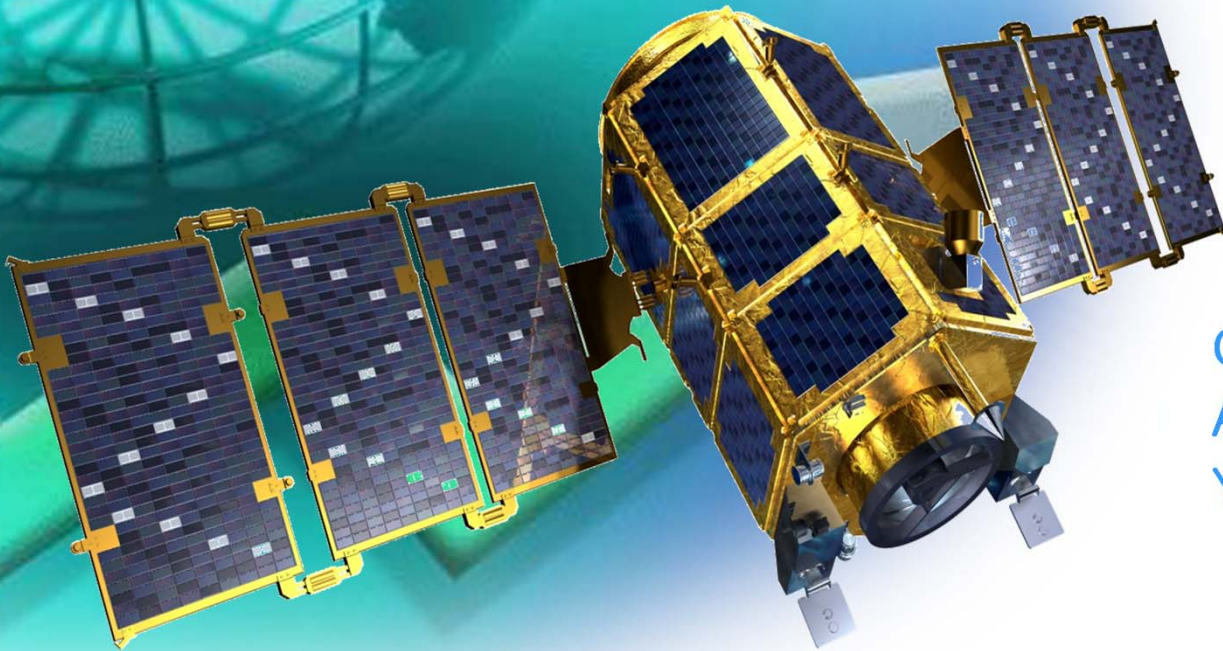
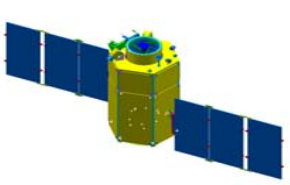


Fall 2011

Spacecraft Systems



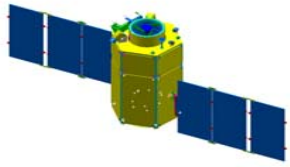
Chandeok Park
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Chapter 11

SPACECRAFT SUBSYSTEMS

- 11.1 Attitude Determination & Control Subsystem
- 11.2 Telemetry, Tracking & Command Subsystem
- 11.3 Command & Data Handling Subsystem
- 11.4 (Electrical) Power Subsystem
- 11.5 Thermal Control Subsystem**
- 11.6 Structures & Mechanism Subsystem
- 11.7 Guidance, Navigation & Control Subsystem



11.5 Thermal Control Subsystem

◆ Thermal Control Subsystem (TCS)

- Maintain all the items of a S/C within their allowed temperature limits during all mission phases using minimum S/C resources
- 2~5% of total S/C cost & of total S/C mass.

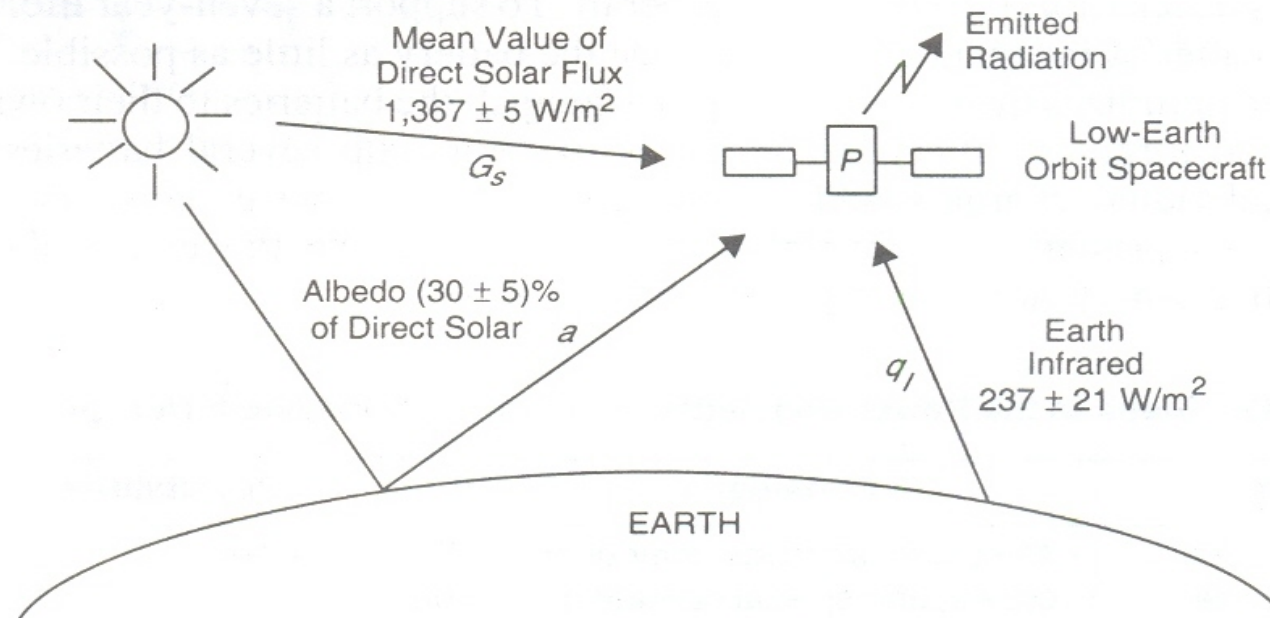
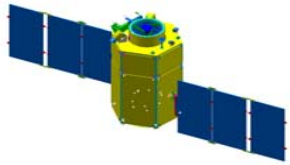


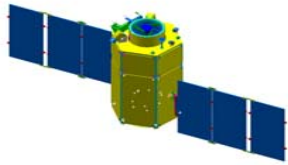
Fig. 11-14. Thermal-Radiation Environment for a Typical Spacecraft. P refers to the power dissipated on board the spacecraft. Values shown are taken from NASA [1982].



11.5 Thermal Control Subsystem

TABLE 11-43. Typical Operating Temperature Ranges for Selected Spacecraft Components. Temperatures in °C.⁽¹⁾

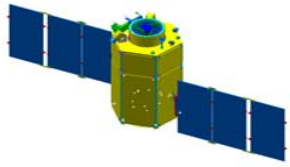
Spacecraft Internal Units			
Worst case envelope ⁽²⁾	0 to +40		
Telecommunications			
Payload units	-10 to +50		
Remote Sensing Payloads			
Optical sensors	+15 to +25	Infrared module ⁽³⁾	-40 to +30
Radiometric units	-10 to +50	Radar units	-10 to +40
Onboard Computer	-10 to +50		
Telemetry & Command units	-10 to +50		
Electrical Power			
Batteries (NiH ₂)	-5 to +20	Batteries (NiCd)	0 to +25
Solar Arrays	-105 to +110	Power control unit	-20 to +55



11.5 Thermal Control Subsystem

Attitude Control			
Sun & Earth sensors	-30 to +50	Magnetometer	-80 to +80
Electronic units	-10 to +55	Momentum and reaction wheels	-5 to +45
Gyro package	0 to +50		
Propulsion			
Propellant tank, filters, valves, lines	+7 to +55	Thrusters	+7 to +65
Harness			
Spacecraft internal	-15 to +55	Spacecraft external	-100 to +100
Thermal Control			
Multilayer Insulation (MLI)	-160 to +250	Heaters, thermostats, heat pipes	-35 to +60
Radiators	-95 to +60		
Structures			
Nonalignment critical	-45 to +65	Alignment critical ⁽⁴⁾	+18 to +22
Mechanisms			
Pyrotechnics	-100 to +120	Electric motors	-45 to +80
Deployment hinge	-45 to +65	Solar array drive assembly	-35 to +60
Antennas			
Parabolic reflector	-160 to +95	TT&C	-65 to +95
GPS antenna	-95 to +70		

- (1) For all equipment units, the reported temperatures apply to their external case or mounting interface. The temperature range for internal components of equipment is wider (e.g. a transistor can operate in the range -30 to +110 °C).
- (2) This is a conservative range envelope used for the spherical spacecraft exercise, cf. Sec. 11.5.5.
- (3) The detector needs a cryogenic system within the module, to maintain the range -200 to -80 °C.
- (4) This range equals the temperature of the integration room where equipment is installed on the spacecraft structure and aligned with a high accuracy. The requirement for the thermal design is simply to maintain the same temperature range also during the space mission. This will assure that on-ground alignments will be maintained (duplicated) also in orbit by limiting the thermo-elastic distortion of the structure.



11.5 Thermal Control Subsystem

◆ 11.5.1 Principles of Heat Transfer

■ Convection

- Thermal energy transfer between a flowing fluid and a solid surface

- $q = hA\Delta T$

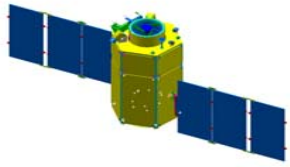
- q : rate of heat flow (W)
- h : heat transfer coefficient ($W/m^2 \cdot K$)
- A : surface area (m^2)
- ΔT : temperature difference between surface and fluid (K)

■ Conduction

- Thermal energy transfer through matter in the absence of fluid motion

- $Q = (kA/\Delta x)(T_1 - T_2)$

- Q : energy transfer rate (W)
- k : thermal conductivity ($W/m \cdot K$)
- A : cross-sectional area (m^2)
- Δx : length of heat transfer path (m)
- T : temperature (K)



11.5 Thermal Control Subsystem

◆ 11.5.1 Principles of Heat Transfer (cont'd)

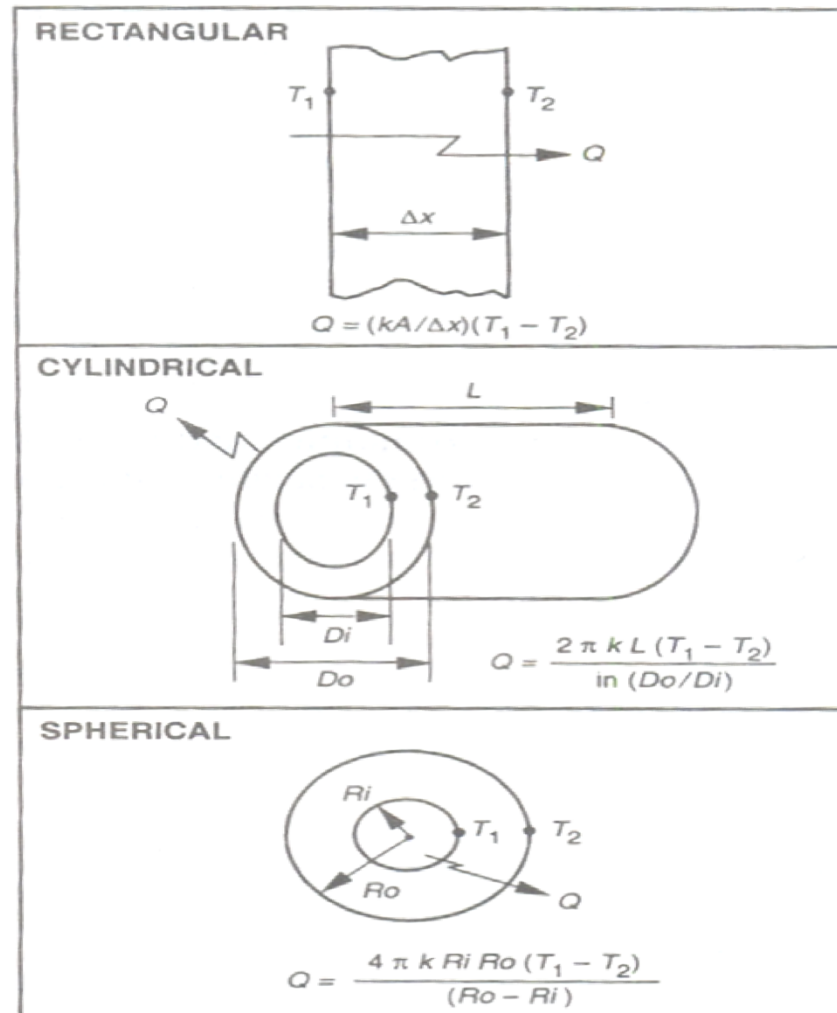
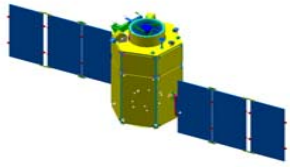


Fig. 11-15. Equations for Steady-State Conduction. These equations help us compute the amount of heat, Q , conducted through each shape, depending on the thermal conductivity, k , of the material, and the other variables.



11.5 Thermal Control Subsystem

◆ 11.5.1 Principles of Heat Transfer (cont'd)

■ Radiation

■ Energy transfer via electromagnetic waves

■ $q = \varepsilon \sigma T^4$

- σ : Stefan-Boltzman Constant ($= 5.605 \times 10^{-8} W/m^2 K^4$)
- ε : emissivity (0~1)
- T : temperature (K)

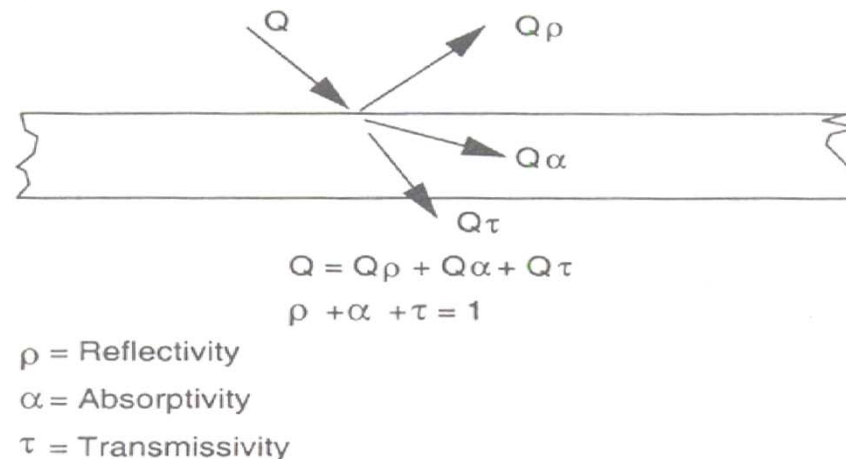
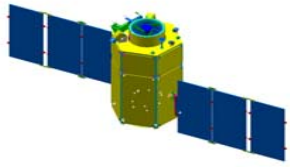


Fig. 11-16. Radiation Properties of Surfaces. Any surface irradiated by electromagnetic radiation absorbs it, α , reflects it, ρ , or transmits it, τ .



11.5 Thermal Control Subsystem

◆ 11.5.1 Principles of Heat Transfer (cont'd)

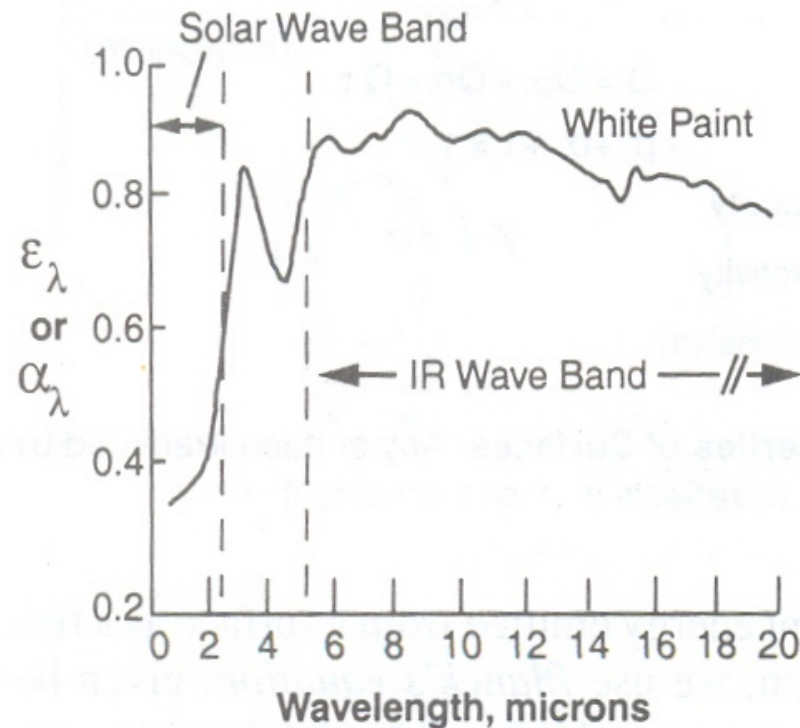
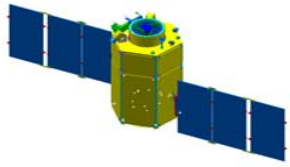


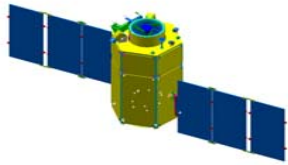
Fig. 11-17. White Paint Emissivity and Absorptivity as a Function of Wavelength. White paint properties from Siegel and Howell [1968]. Low absorptivity in the Solar Wave Band, coupled with high emissivity in the IR Wave Band, means white-painted surfaces stay relatively cool in sunlight.



11.5 Thermal Control Subsystem

◆ 11.5.2 Thermal Control Components

- Passive Thermal Control Subsystem
 - Is generally lighter
 - Requires less electrical power
 - Is less costly than an active subsystem
 - An example is a space radiator thermally coupled to heat-dissipating equipment by conductive paths
- Active Thermal Control Subsystem includes
 - Pumped loop systems
 - Heaters controlled by thermostats
 - Mechanical refrigerators

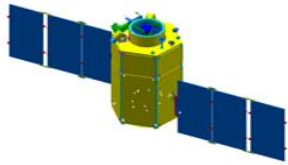


11.5 Thermal Control Subsystem

◆ 11.5.2 Thermal Control Components (cont'd)

TABLE 11-44. Radiation Properties. All values are undegraded beginning of life.

No.	Material	Measurement Temp. (K)	Surface Condition	Solar Absorptivity, α	Infra-red Emissivity ϵ	Absorptivity/Emissivity Ratio	Equilibrium Temp* (°C)
1	Aluminum (6061-T6)	294	As Received	0.379	0.0346	10.95	450
2	Aluminum (6061-T6)	422	As Received	0.379	0.0393	9.64	428
3	Aluminum (6061-T6)	294	Polished	0.2	0.031	6.45	361
4	Aluminum (6061-T6)	422	Polished	0.2	0.034	5.88	346
5	Gold	294	As Rolled	0.299	0.023	13.00	482
6	Steel (AM 350)	294	As Received	0.567	0.267	2.12	207
7	Steel (AM 350)	422	As Received	0.567	0.317	1.79	187
8	Steel (AM 350)	611	As Received	0.567	0.353	1.61	175
9	Steel (AM 350)	811	As Received	0.567	0.375	1.51	168
10	Steel (AM 350)	294	Polished	0.357	0.095	3.76	281
11	Steel (AM 350)	422	Polished	0.357	0.111	3.22	259
12	Steel (AM 350)	611	Polished	0.357	0.135	2.64	234



11.5 Thermal Control Subsystem

◆ 11.5.2 Thermal Control Components (cont'd)

13	Steel (AM 350)	811	Polished	0.357	0.155	2.30	217
14	Titanium (6AL-4V)	294	As Received	0.766	0.472	1.62	176
15	Titanium (6AL-4V)	422	As Received	0.766	0.513	1.49	166
16	Titanium (6AL-4V)	294	Polished	0.448	0.129	3.47	270
17	Titanium (6AL-4V)	422	Polished	0.448	0.148	3.03	251
18	White Enamel	294	Al. Substrate	0.252	0.853	0.30	20
19	White Epoxy	294	Al. Substrate	0.248	0.924	0.27	13
20	White Epoxy	422	Al. Substrate	0.248	0.888	0.28	16
21	Black Paint	294	Al. Substrate	0.975	0.874	1.12	136
22	Silvered Teflon	295		0.08	0.66	0.12	-39
23	Aluminized Teflon	295		0.163	0.8	0.20	-6
24	OSR (Quartz Over Silver)	295		0.077	0.79	0.10	-51
25	Solar Cell-Fused Silica Cover			0.805	0.825	0.98	122

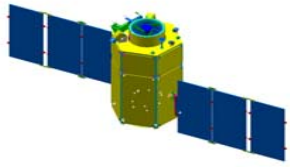
Notes:

Items 1 through 21 are from Brown and Gagola [1965]

Items 22, 23, and 24 are from Ahern, Belcher, et al. [1983]

Item 25 is from NASA Jet Propulsion Laboratory [1976]

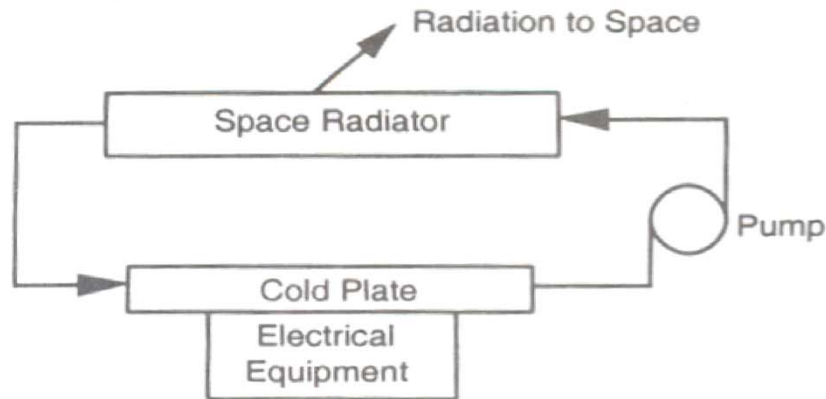
* Temperature of a flat plate located in space (radiation heat exchange with deep space), with a perpendicular solar input of $1,418 \text{ W} \cdot \text{m}^{-2}$, no internal dissipation, and perfect insulation of its backside.



11.5 Thermal Control Subsystem

◆ 11.5.2 Thermal Control Components (cont'd)

ACTIVE THERMAL CONTROL



PASSIVE THERMAL CONTROL

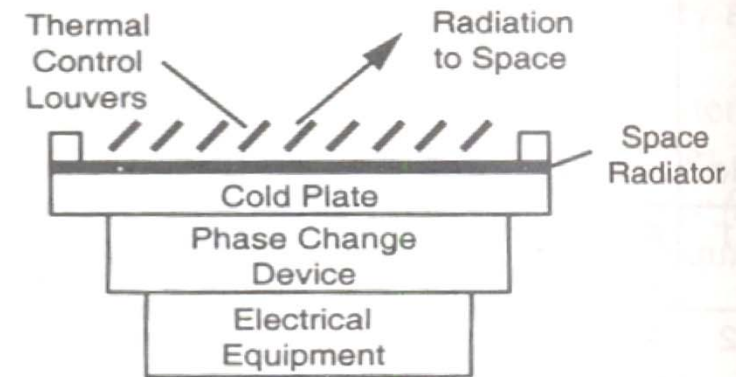


Fig. 11-18. Arrangements for a Spacecraft's Thermal-Control Subsystem.

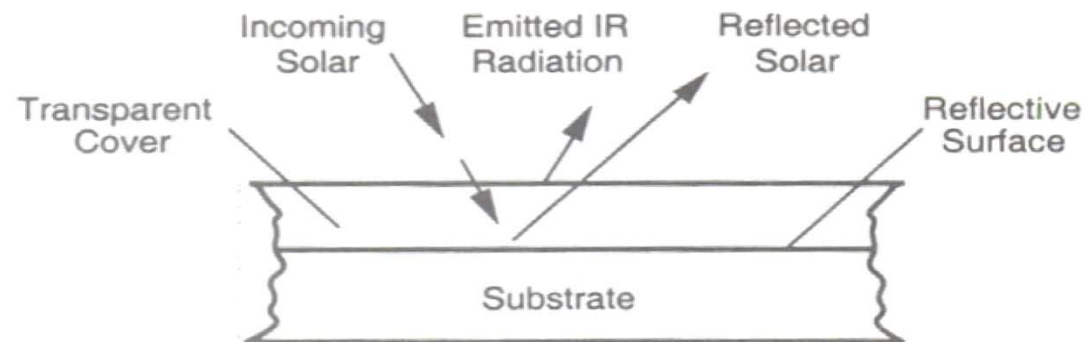
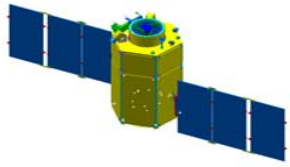


Fig. 11-19. Thermal Performance of the Optical Surface Reflector.



11.5 Thermal Control Subsystem

◆ 11.5.2 Thermal Control Components (cont'd)

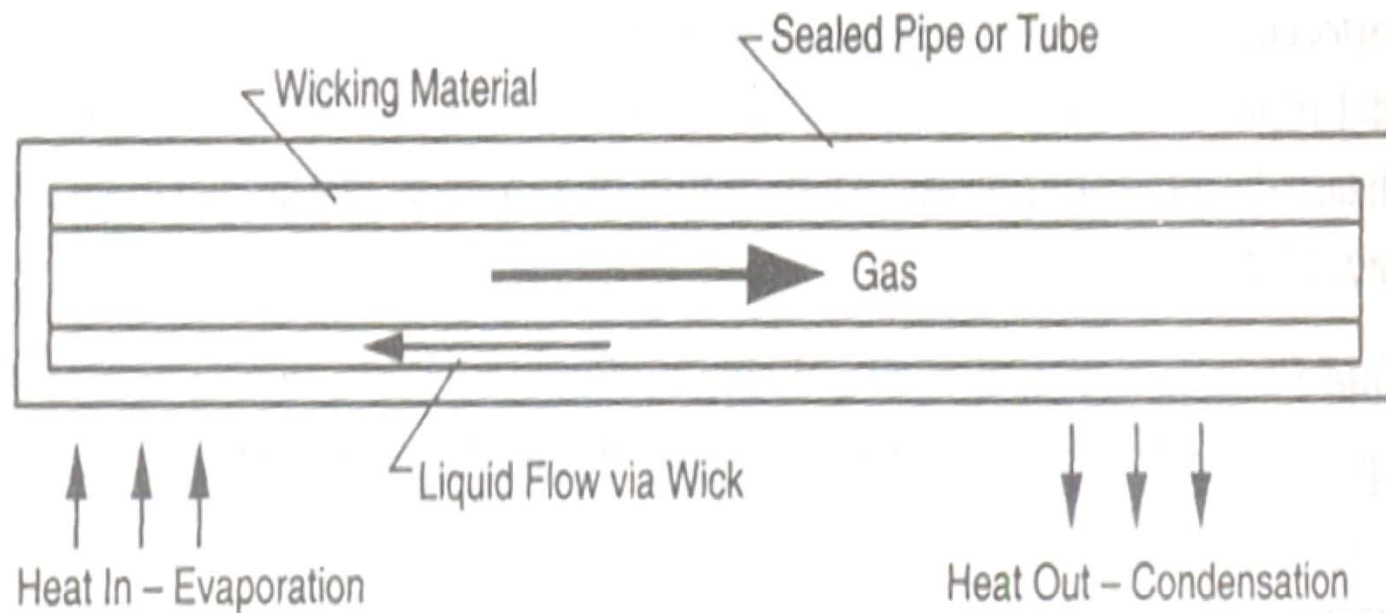
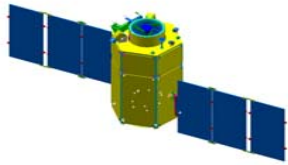


Fig. 11-20. Heat-Pipe Schematic. Heat pipes can transfer up to 1,000 times as much energy as a solid aluminum bar of the same diameter.

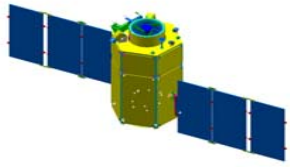


11.5 Thermal Control Subsystem

◆ 11.5.2 Thermal Subsystem Design

TABLE 11-45. The Preliminary Design Process for the Thermal Subsystem of an Isothermal Spacecraft.

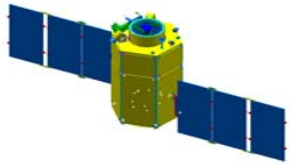
Step	Notes	Where Discussed
1. Determine requirements and constraints	• Identify temperature limits • Payload requirements usually dominate; batteries often have narrow temperature limits • Estimate electrical power dissipation • Use power consistent with worst case hot and worst case cold conditions, if available; otherwise use orbit average power	Table 11-43 Tables 10-8, 10-10, 10-16, 10-19, 10-21, 10-23, 10-30
2. Determine diameter of sphere with surface area equal to surface area of the spacecraft	Make first-order estimates of the spacecraft thermal performance by assuming an isothermal and spherical spacecraft	See example Sec. 11.5.5
3. Select radiation surface property values	Initially assume white paint with $\alpha_s = 0.6$ and $\epsilon_{IR} = 0.8$	Table 11-44, Sec. 11.5.6
4. Compute worst-case hot and cold temperatures for the spacecraft	For upper limit, use high side values for direct solar, albedo, and Earth IR emission; for lower limit use low value for Earth IR emission	Fig. 11-14
5. Compare worst-case hot and cold temperatures (step 4) with temperature limits (step 1)	If worst-case hot temperature is higher than required upper limit, plan to use a deployed radiator with a pumped-looped system. If the opposite is true, use body-mounted radiators	Table 11-46 example



11.5 Thermal Control Subsystem

◆ 11.5.2 Thermal Subsystem Design (cont'd)

limits (step 1)		
6. Estimate required area for body-mounted radiator	Use upper temperature limit for radiator temperature, assume no environmental heat inputs, and use maximum heat dissipation	Eq. 11-23
7. Estimate radiator temperature for worst-case cold conditions	Use the area from step 6 and minimum heat dissipation	Eq. 11-23
8. Determine heater power required to maintain radiator at lower temperature limit	If temperature found in step 7 is less than lower limit, in order to compute the heater power, assume the radiator temperature is at the lower limit and the area is the value defined in step 6	Eq. 11-23
9. Determine if there are special thermal control problems	Identify components with narrow temperature ranges, high power dissipation, or low temperature requirements	Sec. 11.5.2
10. Estimate subsystem weight, cost, and power	If no special problems, use 4.5% of spacecraft dry weight, 4% of the total spacecraft cost, and heater power from step 8	Chap. 20
11. Document reasons for selections	Particularly important to document assumptions	N/A



11.5 Thermal Control Subsystem

◆ 11.5.2 Thermal Subsystem Design (cont'd)

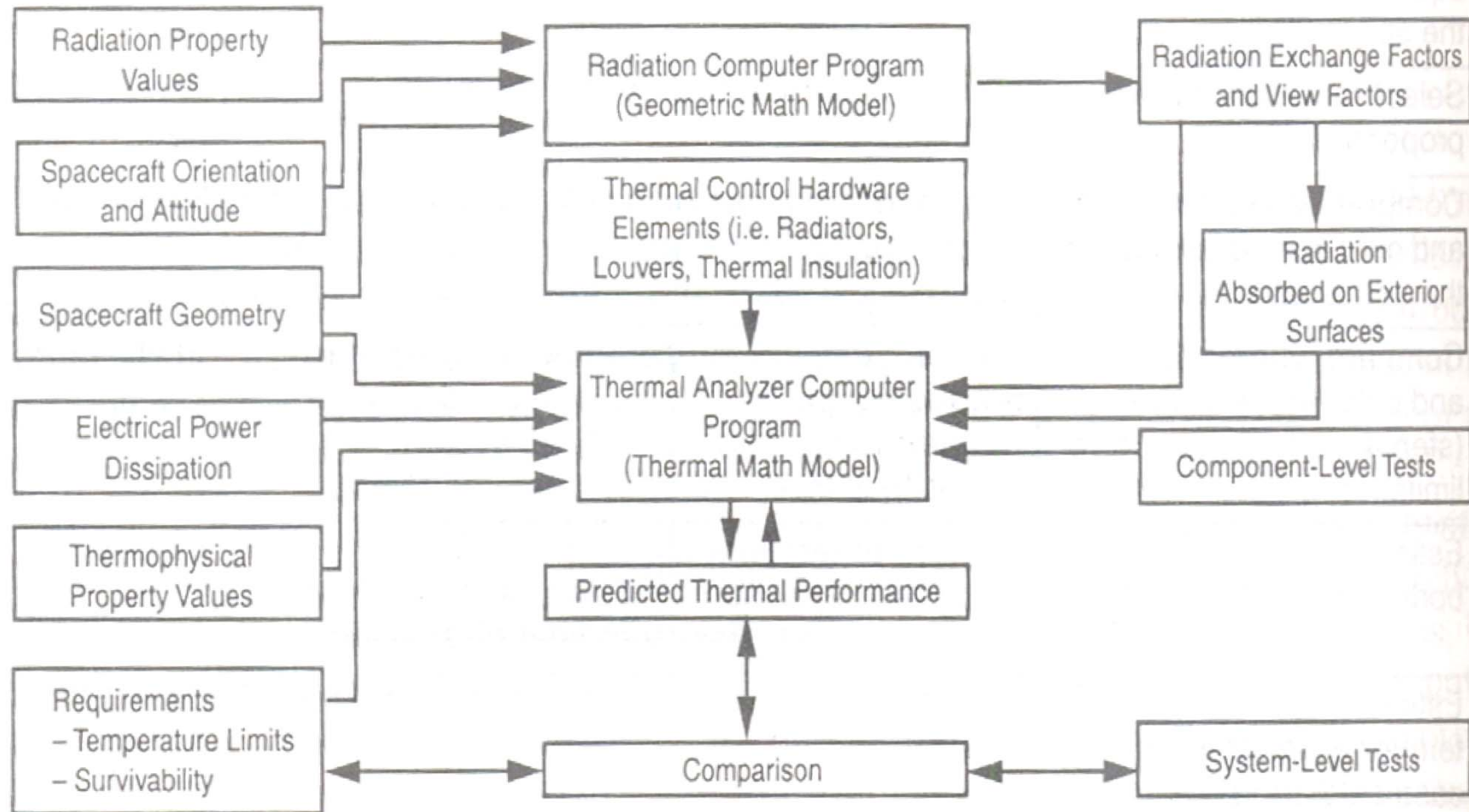
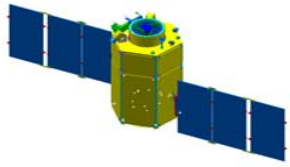


Fig. 11-21. Overall Design Process for Thermal Control of a Spacecraft.



11.5 Thermal Control Subsystem

◆ 11.5.4 Thermal Analysis Concepts

■ Conservation of Energy

- $q_{absorbed} + q_{dissipated} - q_{emitted} = 0$

- $q_{absorbed}$: Reflected Solar Energy, Earth Infrared Emission

- $q_{dissipated}$: S/C Electronic Component

- $q_{emitted}$: Electromagnetic Radiation

- $G_S \alpha \cos \theta + Q_W / A_R - \sigma \varepsilon T^4 = 0$ (11-23)

- G_S : Solar Flux (=1,418 W/m²)

- α : Material's Solar Absorptivity

- θ : Sunlight Incident Angle

- Q_W : Waste Heat rejected by Radiator

- A_R : Radiator Area

- σ : Stefan-Boltzman Constant

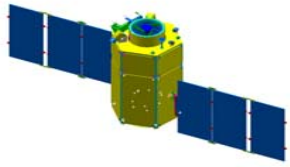
- ε : Infrared Emissivity

- T : Temperature

- Can use (11-23) to:

- Estimate the heat rejection (Q_W), if the radiator operates at a given temperature

- Estimate the required temperature to reject a given amount of heat.

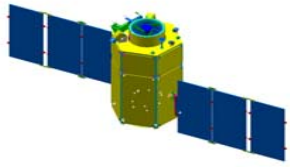


11.5 Thermal Control Subsystem

◆ 11.5.5 Preliminary Design Process

TABLE 11-46. Preliminary Thermal Performance Estimates for the FireSat Spacecraft.

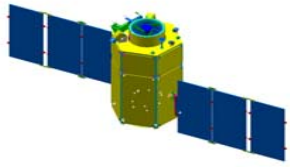
No.	Item	Symbol	Value	Units	Source	Comment
1	Surface area of FireSat	A	13.5	m ²	FireSat	FireSat is 1.5-m cube
2	Diameter of sphere which equals FireSat surface area	D	2.07	m	Spherical geometry	$D = \sqrt{A/\pi}$
3	Max power dissipation	Q_w	170	W	FireSat	Given
4	Min power dissipation	Q_w	80	W	FireSat	Assumed
5	Altitude	H	700	km	Table 7-4	
6	Radius of Earth	R_E	6,378	km	Table 8-2	Rounded to 4 places
7	Angular radius of Earth	ρ	1.1221	rad	Calculated	Defined on page 110
8	Albedo correction	K_a	0.9931	—	Calculated	Defined with Eq. (11-28)
9	Max Earth IR emission at surface	q_l	258	W·m ⁻²	Fig. 11-14	Use for worst-case hot
10	Min Earth IR emission at surface	q_l	216	W·m ⁻²	Fig. 11-14	Use for worst-case cold
11	Direct solar flux	G_S	1,418	W·m ⁻²	Fig. 11-14	Use maximum value



11.5 Thermal Control Subsystem

◆ 11.5.5 Preliminary Design Process (cont'd)

12	Albedo	a	35	%	Fig. 11-14	Use maximum value
13	Emissivity	ϵ	0.8	—	Table 11-44	Assume end of life white paint absorptivity
14	Absorptivity	α	0.6	—	Table 11-44	Assume white paint degrades
15	Worst case hot temperature	T_{MAX}	+27	°C	Eq. 11-34	
16	Worst case cold temperature	T_{MIN}	-87	°C	Eq. 11-35	
17	Upper temperature limit	T_U	35	°C	Table 11-43	Assume 5 °C thermal margin
18	Lower temperature limit	T_L	5	°C	Table 11-43	Assume 5 °C thermal margin
19	Radiator area based on worst case hot conditions	A_R	0.42	m ²	Eq. 11-23	Assume no solar heat input, dissipation of 170 W and radiator temp. of 35 °C
20	Radiator temperature verification for worst case cold conditions	T_R	-18.4	°C	Eq. 11-23	Assume no solar heat input, dissipation of 80 W and area of 0.42 m ²
21	Heater power required to maintain radiator at lower limit	Q_n	33.8	W	Eq. 11-23	Assume no solar heat input, area of 0.42 m ² and radiator temp. of 5 °C

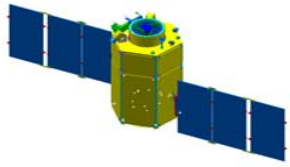


11.5 Thermal Control Subsystem

◆ 11.5.5 Preliminary Design Process (cont'd)

■ Spherical Satellite Analysis Example: FireSat

- I. Establish equipment temperature limits
 - Using Table 11-43, find the typical temperature range for electronics ($0\sim 40^{\circ}\text{C}$) and batteries (NiH2 : $-5^{\circ}\text{C}\sim 20^{\circ}\text{C}$)
- II. Estimate the electrical power dissipation of S/C
 - If possible, determine the power consistent with the worst-case hot & cold condition
 - If NOT possible, use the orbital average value for both conditions
- III. Compute the diameter of a spherical S/C which has a surface area equal to the actual S/C



11.5 Thermal Control Subsystem

◆ 11.5.5 Preliminary Design Process (cont'd)

■ Spherical Satellite Analysis Example: FireSat (cont'd)

IV. Obtain first-order estimates of the S/C thermal performance

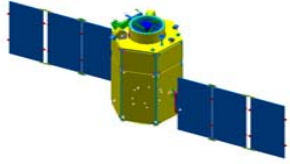
$$T_{MAX,s} = \left[\frac{A_C G_s \alpha + A F q_I \varepsilon + A F G_s a \alpha K_a + Q_w}{A \sigma \varepsilon} \right]^{\frac{1}{4}} \quad (11-34)$$

$$T_{MIN,s} = \left[\frac{A F q_I \varepsilon + Q_w}{A \sigma \varepsilon} \right]^{\frac{1}{4}} \quad (11-35)$$

$$\begin{cases} T_{MAX,s} = 27^\circ C \\ T_{MIN,s} = -87^\circ C \end{cases} \quad \text{for FireSat}$$

Consult Table 11-46 'Preliminary Thermal Performance Estimates' for FireSat

- Worst-Case Hot
 - Electronics are O.K.
 - Battery needs thermal radiation
- Worst-Case Cold
 - Needs thermal control



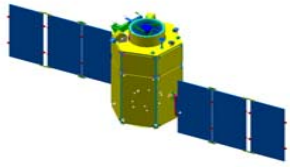
11.5 Thermal Control Subsystem

◆ 11.5.5 Preliminary Design Process (cont'd)

- Spherical Satellite Analysis Example: FireSat (cont'd)
 - v. Assess the performance of body-mounted radiator

$$G_s \alpha \cos \theta + Q_w / A_R - \sigma \epsilon T^4 = 0 \quad (11-23)$$

- Assume worst-case hot temperature to be 40°C for electronic equipment
 - Set margin to be 5°C, then $T = 40 - 5 = 35^\circ\text{C}$ (separate consideration for battery)
 - $Q_{w,max} = 170W \Rightarrow (11-23) \Rightarrow A_R = 0.42m^2$ (radiator area)
 - Heat dissipation (isolate battery)
- Worse-Case cold conditions
 - $Q_{w,min} = 80W, A_R = 0.42m^2 \Rightarrow (11-23) \Rightarrow$ radiator temperature: $-18.4^\circ\text{C} < 5^\circ\text{C}$ (Margin= 5°C)
- Therefore, the radiator will require:
 - Thermal Louver to modulate the heat flow
 - Thermostat and Heater to maintain the temperature above the lower limit
- Heat Power required to maintain radiator at lower limit
 - $A_R = 0.42m^2, T = 5^\circ\text{C}, (11-23) \Rightarrow Q = 33.8W$
 - This power at the radiator is in addition to the 80W dissipated from the electronic equipment

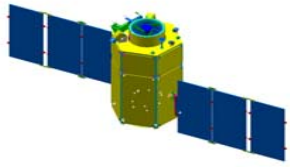


11.5 Thermal Control Subsystem

◆ 11.5.6 Thermal Control Design Consideration

TABLE 11-47. Specific Heat Capacities of Materials. All data are at room temperature. (From various sources.)

Material	Specific Heat Capacity— c_p (J·kg ⁻¹ K ⁻¹)	Material	Specific Heat Capacity— c_p (J·kg ⁻¹ K ⁻¹)
Copper	390	Nylon	1,680
Aluminum Alloys	920	Carbon Fiber	840
Stainless Steel	503	Air (1 atm.)	1,006
Beryllium	1,800	Water	4,182
Magnesium	1,050	Hydrazine (liquid)	3,100
Titanium	520	Gallium Arsenide	335
Nickel	460	Ceramics	800
Tungsten	142	Silicon	712
Teflon	1,006	Kapton	1,006

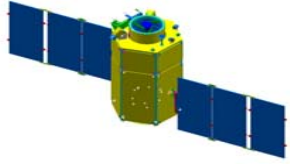


11.5 Thermal Control Subsystem

◆ 11.5.6 Thermal Control Design Consider. (cont'd)

TABLE 11-48. Thermal Conductivities. (From Sandia National Laboratory [1981] and Gilmore [1994]). This table provides additional properties of materials.

Material	Conductivity (W·m ⁻¹ ·K ⁻¹)	Material	Conductivity (W·m ⁻¹ ·K ⁻¹)
Copper	389	Ethylene Glycol	2.42
Aluminum Alloy 2017	164	Water	0.536
Aluminum Alloy 3003	156	Silicone	0.151
Aluminum Alloy 308	96.9	Perlite	0.0692
Aluminum Alloy 2219-0	172.9	Calcium Silicate	0.0548
Aluminum Alloy 6061-T6	167.7	Mineral Fiber Board	0.9548
Aluminum Alloy 7075-T6-T7	121	Mineral Fiber Block	0.0360
Steel 17-4PH	18	Phenolic	0.0332
Inconel 718	11.2	Glass Fiber Block	0.0317
Magnesium AZ31B H24	76.1	Urea Formaldehyde	0.0317
Titanium Ti-6Al-6V	7.26	Polystyrene	0.0288
Beryllium	150.4	Air	0.0260
Plain Carbon Steel	51.9	Polyurethane	0.0231
Stainless Steel Type 304	17.3		



11.5 Thermal Control Subsystem

◆ 11.5.6 Thermal Control Design Consider. (cont'd)

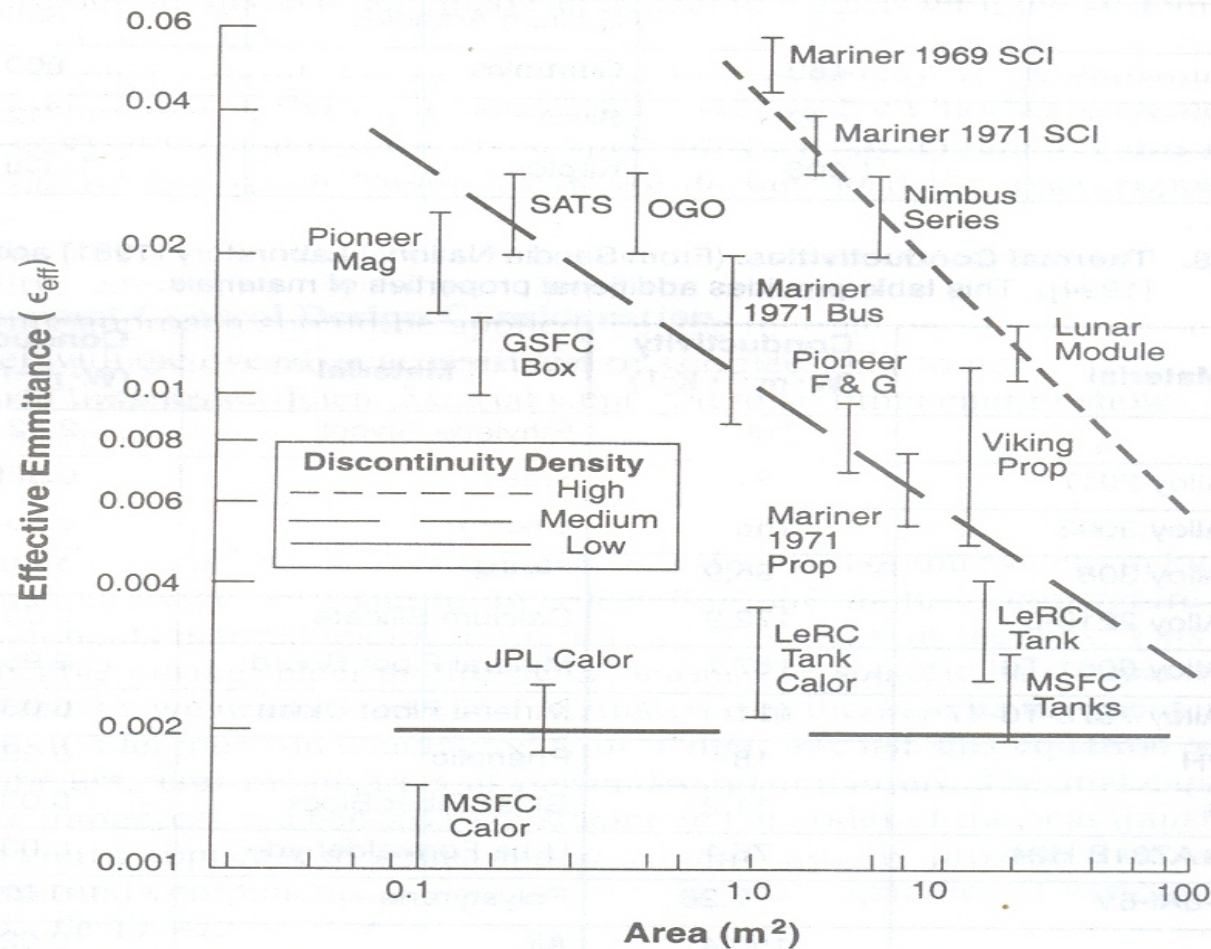
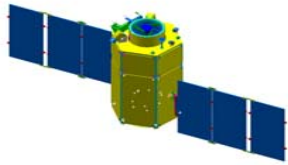


Fig. 11-22. Performance of Multilayer Insulation. In this figure SCI stands for science scan platform, Mag for magnetometer, Prop for propellant, Calor for calorimeter, and F & G for the Pioneer Spacecraft series designations (Stimpson and Jaworski [1972]).



11.5 Thermal Control Subsystem

◆ 11.5.6 Thermal Control Design Consider. (cont'd)

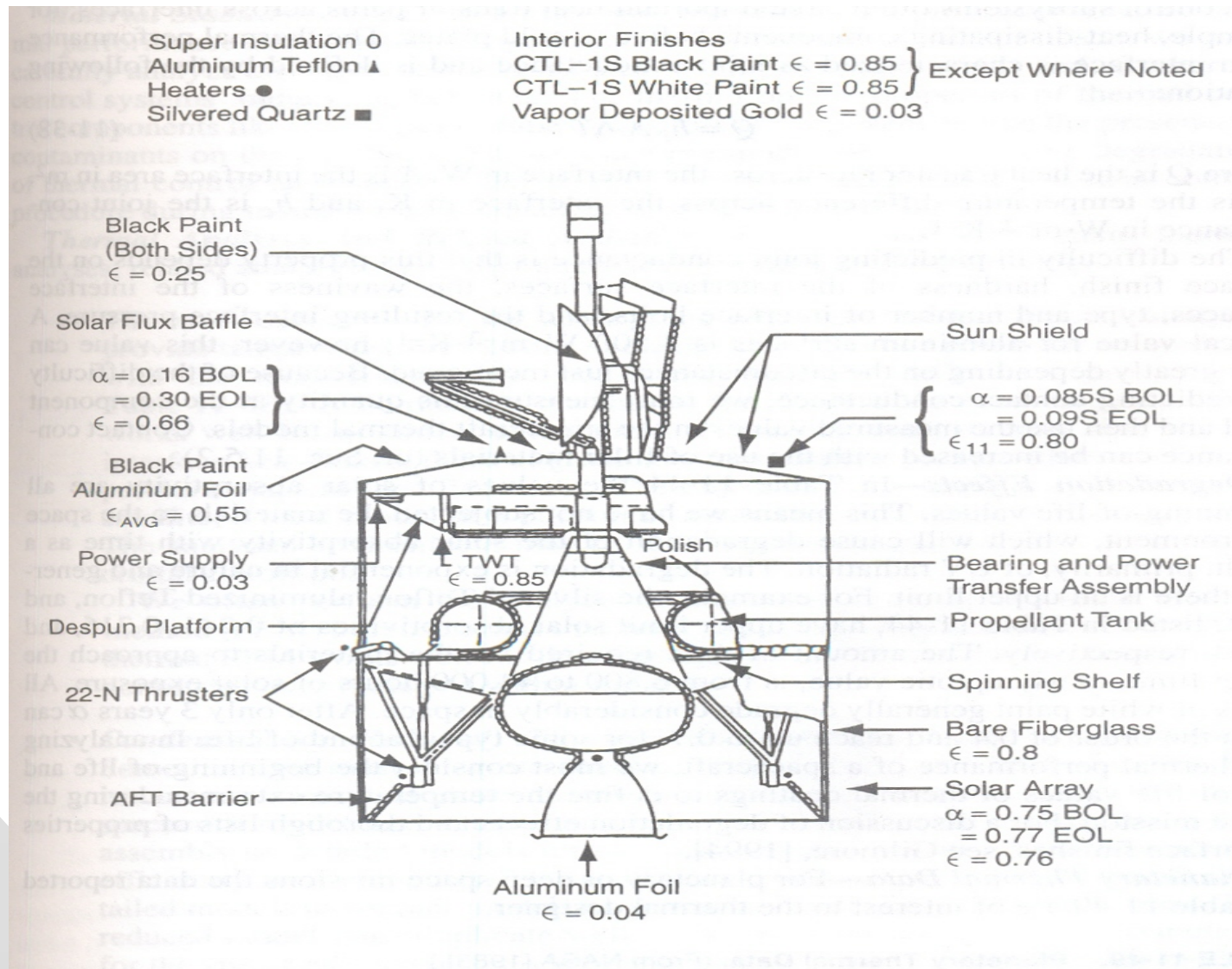
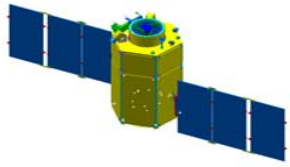


Fig. 11-23. Typical Design Practice for Thermal Control.



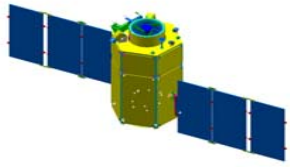
11.5 Thermal Control Subsystem

◆ 11.5.6 Thermal Control Design Consider. (cont'd)

TABLE 11-49. Planetary Thermal Data. (From NASA [1983].)

Planet	Maximum Solar Radiation Intensity (W·m ⁻²)*	Average Albedo
Mercury	9,350	0.06
Venus	2,700	0.61
Earth	1,418	0.34
Moon	1,418	0.07
Mars	600	0.15
Jupiter	52	0.41
Saturn	15	0.42
Uranus	4	0.42
Neptune	1.5	0.52
Pluto	0.9	0.16

*Rounded up figures.



11.5 Thermal Control Subsystem

◆ 11.5.7 Thermal Testing

- Perform thermal tests at component, subsystem and system levels to verify the thermal design and to demonstrate the proper operation

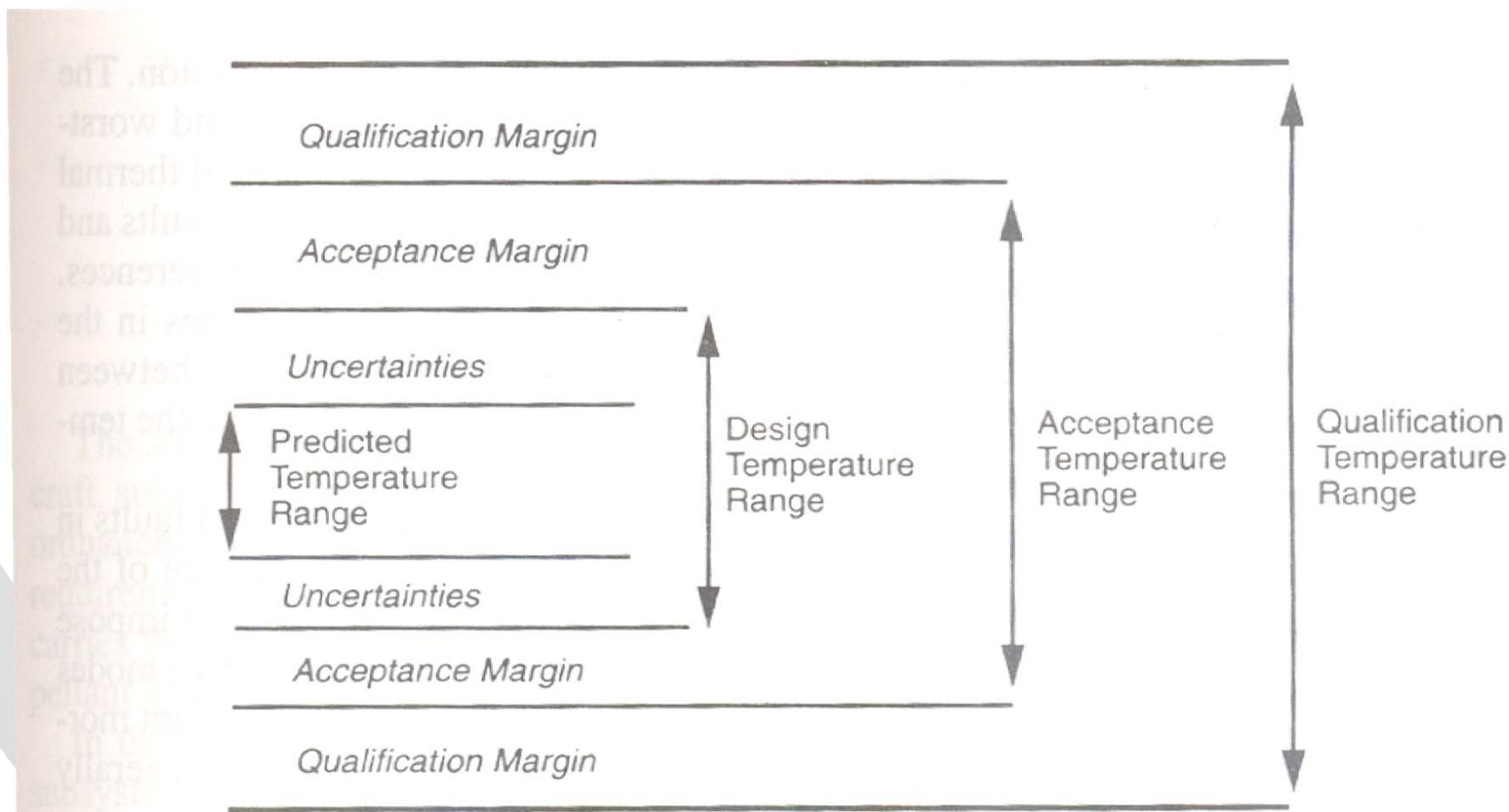


Fig. 11-24. Temperature Definitions for TCS.